ANLP-UniSo at ArGuard 2026 Task B: Multi-Task MARBERT with Focal Loss and Hierarchical Consistency for Arabic Harmful Prompt Detection

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Abstract

Detecting harmful prompts directed at large language models is a critical safety challenge, especially in Arabic where dialectal variation, code-switching, and culturally specific expressions of harm complicate classification. The ArGuard 2026 shared task addresses this problem through two subtasks: binary classification of prompts as safe or unsafe (B1) and fine-grained classification of unsafe prompts into seven harm domains (B2). We participate with a multi-task fine-tuning of MARBERT that shares encoder representations across a binary head and a seven-class granular head. Training combines focal loss with class weights and label smoothing, a hierarchical consistency regularizer that discourages contradictory safe/unsafe and granular predictions, a weighted sampler, and light character-level augmentation. On the official development set the system reaches a binary macro-F1 of 0.9618 and a granular macro-F1 of 1.0000; on the blind test set it obtains binary macro-F1 0.688 and granular macro-F1 0.752. We analyse the substantial development–test gap and discuss limitations of the submitted system.

1 Introduction

Large language models (LLMs) are increasingly deployed in Arabic-speaking contexts. Harmful user prompts covering adult content, self-harm, hate speech, fraud, harassment, bullying, and harm to others pose safety risks. Detecting such prompts before they reach an LLM is therefore a practical safeguard (Alam et al., 2026).

ArGuard 2026 Task B (Alam et al., 2026) addresses this need with two subtasks:

- **B1 (Binary):** Classify an Arabic prompt as *safe* or *unsafe*.
- **B2 (Granular):** Assign an unsafe prompt to one of seven domains: adult content, bullying, fraud or illegal activities, harassment, harm to others, hate speech, and self-harm.

Arabic introduces additional challenges: dialectal variation (MSA and multiple regional dialects), code-switching, sarcasm, and culturally specific expressions of harm. We participate with a single multi-task system based on MARBERT (Abdul-Mageed et al., 2021), a transformer pre-trained on a large collection of Arabic tweets that covers both MSA and dialectal Arabic.

Our contributions are: (i) a multi-task architecture jointly optimizing binary and granular objectives; (ii) focal loss, class weighting, label smoothing, and hierarchical consistency regularization; and (iii) light character-level augmentation and weighted sampling.

2 Related Work

Research on harmful and unsafe Arabic content has evolved from task-specific detection of abusive and offensive language toward broader content-safety classification. Early work by Mubarak et al. (2017) investigated abusive language detection on Arabic social media, while the *L-HSAB* dataset addressed hate speech and abusive language in Levantine Arabic, highlighting the challenges introduced by dialectal variation (Mulki et al., 2019). Shared tasks such as OSACT4 and OSACT5 subsequently established standardized benchmarks for Arabic offensive language and hate speech detection (Mubarak et al., 2020, 2022). More recent work has expanded the scope to other harmful behaviors, including cyberbullying (Allwaibed et al., 2025) and toxicity detection in Arabic social media (Hatem et al., 2025).

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Transformer-based models and multi-task learning have increasingly been adopted for Arabic harmful-content classification. Bennessir et al. (2022) explored transformer-based and multi-task approaches for Arabic hate speech detection, including MARBERT-based representations and task-specific layers. AlKhamissi and Diab (2022) further proposed a MARBERT-based multi-task framework with self-correction for offensive language, hate speech, and fine-grained hate speech classification. Similarly, Alrashidi et al. (2023) investigated multi-task learning with MARBERT for multi-aspect Arabic abusive-content classification. More recently, Ben Arbia et al. (2026) systematically compared classical machine learning and transformer-based approaches for Arabic politeness detection at AdabEval 2026. Their best system used MARBERT with Focal Loss to address substantial class imbalance, substantially outperforming classical machine learning baselines. This result further supports the effectiveness of MARBERT and imbalance-aware objectives for challenging Arabic text classification tasks.

With the emergence of large language models (LLMs), recent research has shifted toward the broader problem of Arabic content safety. AraSafe introduced a benchmark for Arabic LLM safety with binary and fine-grained safety categories (Mubarak et al., 2025), while ArabicDialectSafety emphasized dialect-aware evaluation across multiple Arabic varieties (Zaghouani et al., 2026). ArGuard 2026 extends this direction to harmful Arabic prompts submitted to LLMs, with a binary safe/unsafe task and a fine-grained classification task covering seven harm domains (Alam et al., 2026).

Our work focuses on ArGuard Task B using a multi-task MARBERT architecture for joint binary and fine-grained harm classification. Building on prior work with MARBERT and imbalance-aware learning (Abdul-Mageed et al., 2021), we combine focal loss, class weighting, label smoothing, hierarchical consistency regularization, and light character-level augmentation.

3 Data and Preprocessing

We use the official ArGuard Task B data (Alam et al., 2026), consisting of training, labeled development, unlabeled development, and blind test splits. After removing rows with missing labels, the training and labeled development sets contain 16,831 and 2,405 prompts, respectively. Approximately 80% of training prompts are unsafe; adult content is the most frequent granular class, while hate speech is the least frequent.

The data cover MSA, Syrian, Moroccan, Egyptian, Palestinian, and Algerian varieties. Figure 1 summarizes the binary, granular, and dialect distributions. No external labeled datasets were used.

Split design. We use the official partitions without re-splitting or re-sampling. Because no independent validation fold is provided, the labeled development set is used for both checkpoint selection and hyper-parameter tuning and for reporting development metrics. This introduces selection bias and may contribute to the development–test gap discussed in Section 6.

Preprocessing is minimal: The original Arabic prompts are preserved, with lowercasing only where required for label normalization. Training optionally uses character-level augmentation through random space insertion or character duplication with probability 0.25. We also experimented with multi-dialect knowledge injection but disabled it after observing label leakage; the final model receives only the raw prompt.

4 System Architecture and Training

4.1 Model

We fine-tune MARBERT (`UBC-NLP/MARBERT`) end-to-end. A shared linear projection with GELU and dropout (0.2) is applied to the `[CLS]` representation. Two linear heads produce binary logits (2 classes) and granular logits (7 classes). Figure 2 gives an overview of the full system. The model is trained with automatic mixed precision on a single NVIDIA T4 GPU (approximately 15.6 GB).

4.2 Objectives

The total loss is a weighted combination of three terms:

$$\mathcal{L} = 0.4\,\mathcal{L}_{\text{bin}} + 0.5\,\mathcal{L}_{\text{gran}} + 0.1\,\mathcal{L}_{\text{hier}}, \qquad (1)$$

where $\mathcal{L}_{\text{bin}}$ and $\mathcal{L}_{\text{gran}}$ are focal losses (Lin et al., 2017) with $\gamma = 2.0$, inverse-frequency class weights, and label smoothing of 0.08. The hierarchical term encourages consistency: when the binary head assigns high probability to the safe class, the maximum softmax probability over the

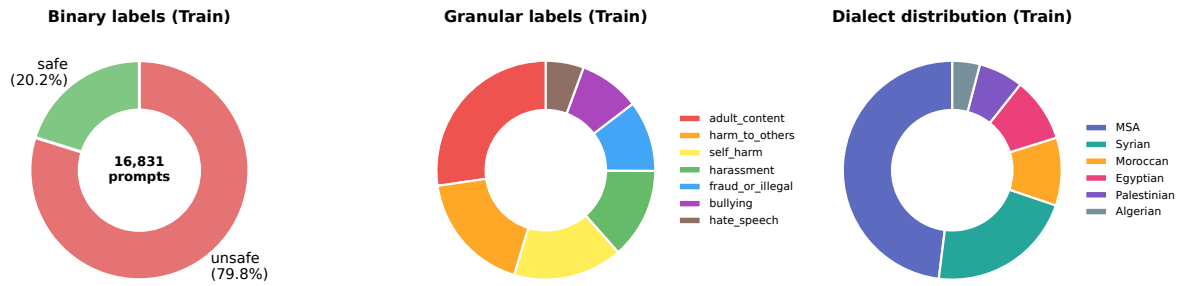


Figure 1: Training-set distributions: binary labels (left), seven granular harm domains (centre), and dialect tags (right).

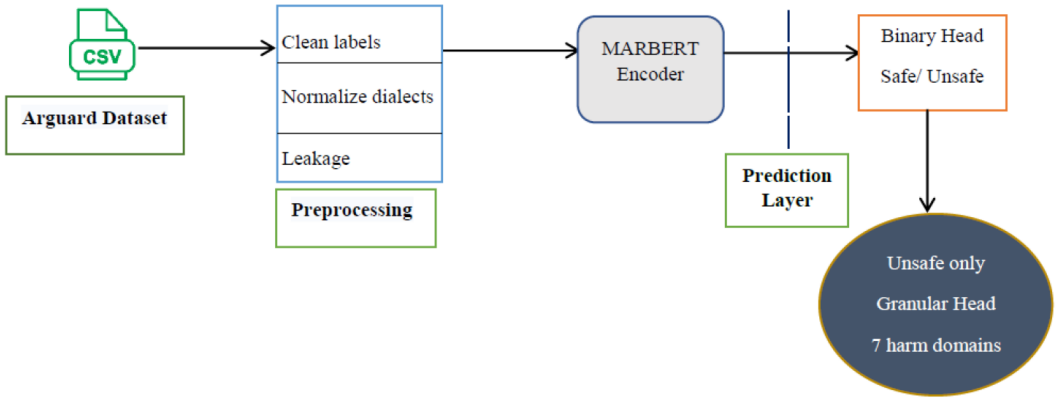


Figure 2: Overview of the multi-task classification system.

seven granular (harm) classes should be low:

$$\mathcal{L}_{\text{hier}} = \alpha \cdot \mathbb{E}\left[p(\text{safe}) \cdot \max_k p(\text{granular}_k)\right], \quad \alpha = 0.25. \tag{2}$$

4.3 Optimization

We use AdamW (lr $= 2 \times 10^{-5}$, weight decay 0.01), a cosine schedule with 8% warm-up, batch size 32, maximum sequence length 128, and six epochs. A weighted random sampler balances the binary classes during training. Gradient clipping at norm 1.0 is applied. The checkpoint with the highest combined development score (average of binary and granular macro-F1) is retained for inference.

5 Experimental Results

Table 1 reports official development and blind-test metrics. On development, the model achieves binary macro-F1 of 0.9618 and granular macro-F1 of 1.0000. On the blind test set, the scores decrease to 0.688 and 0.752, respectively.

The development–test gap is substantial, with macro-F1 decreases of approximately 0.27 for binary classification and 0.25 for granular classification. Figure 3 shows the development confusion matrices. The most difficult granular classes are *harm_to_others*, *bullying*, and *harassment*.

Split	Task	Acc.	P	R	Macro-F1
Dev	Binary	0.976	–	–	0.962
	Granular	1.000	–	–	1.000
Test	Binary	0.706	0.778	0.709	0.688
	Granular	0.768	0.762	0.756	0.752

Table 1: Development and official test results for team `anlp-uniso`.

Figure 3 shows row-normalized confusion matrices on the labeled development set. Binary errors are rare in both directions. On the granular task, the hardest classes are *harm_to_others*, *bullying* and *harassment*, which are frequently confused with one another.

6 Error Analysis and Development–Test Gap

The large development–test gap likely results from several factors. First, the labeled develop-

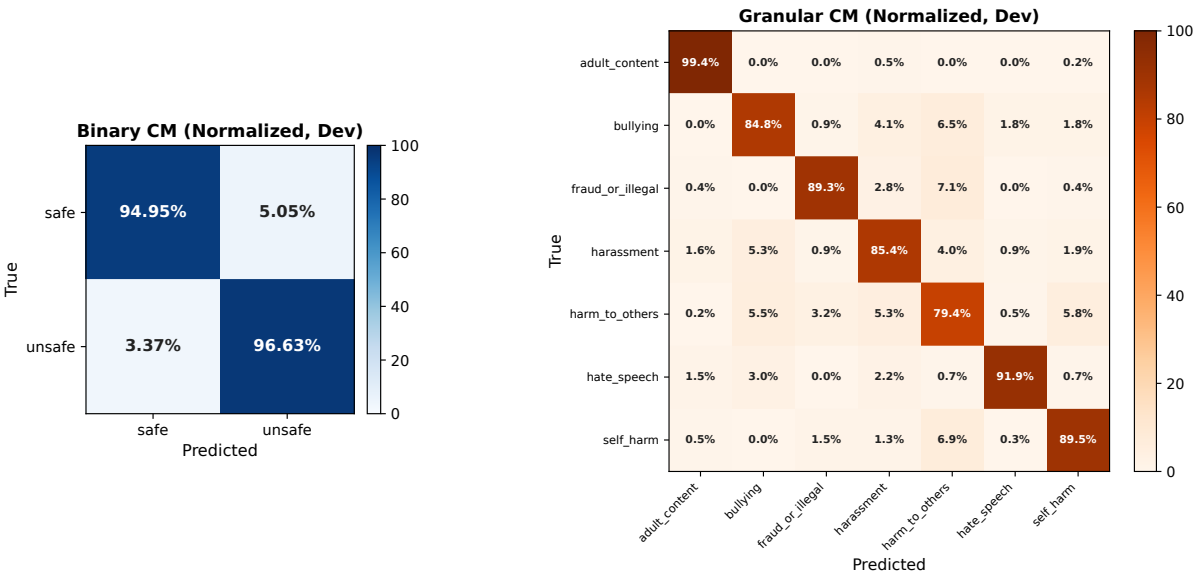


Figure 3: Row-normalized confusion matrices on the labeled development set. Left: binary classification; right: seven-way granular classification.

ment set was used for checkpoint selection, hyperparameter tuning, and evaluation because no independent validation fold was provided, leading to optimistic development estimates.

Second, an earlier pipeline injected gold granular descriptions into the input. This mechanism was removed after detecting label leakage; however, residual artifacts may have contributed to the unusually high development score. Thus, the granular development macro-F1 of 1.000 should be interpreted cautiously.

Third, dialect and distribution shifts may affect generalization. Training is dominated by MSA and Levantine varieties, while Moroccan and Algerian prompts and short or implicit formulations are less represented. Finally, overlapping semantic boundaries between harassment, bullying, and hate speech may further reduce robustness under test-time shift.

7 Conclusion

We presented a multi-task MARBERT system for Arabic harmful-prompt detection that jointly performs binary safety and seven-way harm classification. Focal loss, class weighting, hierarchical consistency, and light augmentation provide strong development performance, but the blind test results reveal a substantial generalization gap. Future work will explore leakage-free knowledge injection, dialect-specific adaptation, and independent validation protocols.

Limitations

The system relies only on official ArGuard data and was trained on a single GPU. The development–test gap indicates selection bias and distribution shift, while character-level augmentation provides limited dialectal variation. Future work should evaluate larger models, dialect-aware adaptation, and more robust validation protocols.

Ethical Considerations

Harmful-content classifiers can be misused for over-censorship or can fail to protect vulnerable users if they under-detect certain categories (especially self-harm). All training data were obtained from the official shared-task release and handled only for research purposes. Sensitive examples were never displayed outside the experimental pipeline. We encourage deployment of such systems only as one component of a broader safety stack that includes human oversight, appeal mechanisms, and continuous monitoring for demographic and dialectal bias.

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